

Total No. of Printed Pages: 4

SUBJECT CODE NO: - RNP-8007
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - II
Examination November / December 2025 (To be held in February-2026)
Applied Mathematics-II

[Time: 3 Hours]

[Max. Marks: 60]

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed.
- 2) Q.no.1 is compulsory.
- 3) Solve any four questions from Q.no's 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve the following questions

20

1. Bowley's skewness uses which central value?

- a) Mean
- b) Mode
- c) Median
- d) SD

2. If mean = 50 and SD = 0, data is:

- a) Spread
- b) Constant
- c) Skewed
- d) Insignificant

3. If mode is not available, Pearson's skewness =

- a) $\frac{3(\text{Mean} - \text{Median})}{\text{SD}}$
- b) $\frac{3(\text{Mean} - \text{Median})}{\text{Variance}}$
- c) $\frac{3(\text{Mode} - \text{Median})}{\text{SD}}$
- d) None

4. For the data: 3, 7, 7, 8, 9, mode is:

- a) 3
- b) 7
- c) 8
- d) 9

5. The equation of tangents to the curve at origin represented by the equation y^2

$(2a-x)=x^3$ is

- a) $x = 0, y=0$
- b) $x = a, y = 2a$
- c) $x = 0, y=0$
- d) $x=2a, y=a$

6. The expression for arc length in parametric form is

$$\begin{aligned} \text{a) } ds &= \int_a^b xy \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dsb \, ds = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ \text{c) } ds &= \int_a^b x \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dx \, ds = \int_a^b y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \end{aligned}$$

7. The value of $\int_0^\infty e^{-x} x^{n-1} dx \, n > 0$ is called as

- a) Beta function
- b) Gamma function
- c) Exponential function
- d) Logarithmic function

8. The value of $\int_0^{\frac{\pi}{2}} \sin^4 \theta \cos^6 \theta \, d\theta$ is equal to

- a) $\frac{3}{512}$
- b) $\frac{512}{3}$
- c) $\frac{3\pi}{512}$
- d) $\frac{512\pi}{3}$

9. The double integral $\iint f(x, y) \, dA$ represents:

- a) Line integral
- b) Area under curve
- c) Volume under surface
- d) None

10. The value of $\int_0^1 \int_0^y dx \, dy$ is

- a) 1
- b) $\frac{1}{8}$
- c) 0
- d) $\frac{1}{2}$

11. Area of cardioid $r=1-\cos \theta$ is:

- a) 2π
- b) 3π
- c) π
- d) 4π

12. Volume bounded by $x = 0$, $y = 0$, $z=x+y$ and plane $z = 0$ is:

- a) Zero
- b) Non-zero
- c) Infinity
- d) Undefined

SECTION B

Q2

a) Calculate standard Deviation of the following frequency distribution

5

Weekly expenditure below	5	10	15	20	25
No of students	6	16	28	38	46

5

b) Find Karl-Pearson's coefficient of skewness for the given distribution:

X:	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
F:	2	5	7	13	21	16	8	3

Q3 a) Evaluate $\iint_A xy(x+y) dx dy$ where A is the area between x^2 & $y = x$ **5**

b) Change to polar co-ordinate & evaluate $\int_0^{2a} \int_0^{\sqrt{2x-x^2}} \frac{xdxdy}{x^2+y^2}$ **5**

Q4 a) Evaluate: $x\sqrt{2ax-x^2} dx$ **5**

b) Prove that $\beta(m, n) = \int_0^1 \frac{t^{m-1}}{(1-t)^{m+n}} dt$ **5**

Q5 a) Trace the curve $r = a(1 + \sin \theta)$ with full justification **5**

b) Trace the curve $r = a \sin 2\theta$ with justification **5**

Q6 a) Evaluate: $\int_{-1}^1 \int_0^z \int_{x-z}^{x+z} (x+y+z) dx dy dz$ **5**

b) Evaluate $\iiint z^2 dz dx dy$ over the volume bounded by the surfaces $x^2 + y^2 = a^2$ & $x^2 + y^2 = z, z = 0$ **5**

Q7 a) Find the Q.D. and its co-efficient **5**

C.I.	20-25	25-30	30-35	35-40	40-45	45-50
f	3	8	16	22	15	5

b) Calculate mean deviation about mean **5**

C.I.	0-10	10-20	20-30	30-40	40-50	50-60	60-70
f	6	5	8	15	7	6	3